8.2 Integration by Parts

Integration by parts is a technique for simplifying integrals of the form $\int f(x)g(x)dx$.
It is useful when f can be differentiated repeatedly and g can be integrated repeatedly without difficulty. Integration by parts can be applied to integrals like $\int x\cos x\,dx$, $\int x^2e^x dx$, $\int e^x\cos x\,dx$ and $\int \ln x dx$.

Product Rule in Integral Form

If u and v are differentiable functions of x, the Product Rule says that

$$\frac{d}{dx}[\,u(x)\,v(x)\,] = u'(x)\,v(x) + u(x)\,v'(x).$$

In terms of indefinite integrals, this equation becomes

$$\int \frac{d}{dx}[\,u(x)\,v(x)\,]\,dx = \int [\,u'(x)\,v(x) + u(x)\,v'(x)]\,dx$$

or

$$\int \frac{d}{dx}[\,u(x)\,v(x)\,]\,dx = \int u'(x)\,v(x)\,dx + \int u(x)\,v'(x)\,dx.$$

Rearranging the terms of this last equation, we get

$$\int u(x)\,v'(x)\,dx = \int \frac{d}{dx}[\,u(x)\,v(x)\,]\,dx - \int v(x)\,u'(x)\,dx,$$

leading to the following **integration by parts formula**.

Integration by Parts Formula	$\int u(x)v'(x)dx = u(x)v(x) - \int v(x)u'(x)dx$	(1)

The formula is often given in differential form. With $v'(x)dx = dv$ and $u'(x)dx = du$, the integration by parts formula becomes

Integration by Parts Formula – Differential Version	$\int udv = uv - \int vdu$	(2)

This formula expresses one integral $\int udv$, in terms of a second integral, $\int vdu$. With a proper choice of u and v, the second integral may be easier to evaluate than the first. In using the formula, various choices may be available for u and dv (see Example 1).

Example 1: Find $\int x\cos x\,dx$.

Solution:

(1) Let $u = x$ and $dv = \cos x\,dx$. Then $du = dx$ and $v = \sin x$.

$$\int x\cos x\,dx = \int udv = uv - \int vdu = x\sin x - \int \sin x dx = x\sin x + \cos x + C.$$

Remark:
There are four choices available for $u(x)$ and $v'(x)$ in Example 1:

1. Let $u(x) = 1$ and $v'(x) = x\cos x$.
2. Let $u(x) = x$ and $v'(x) = \cos x$.
3. Let $u(x) = x\cos x$ and $v'(x) = 1$.
4. Let $u(x) = \cos x$ and $v'(x) = x$.

We used choice 2 in Example 1. The other three choices lead to integrals that we do not know how to evaluate. For instance, Choice 3, with $u'(x) = \cos x - x\sin x$, leads to the integral

$$\int (x\cos x - x^2\sin x)\, dx.$$

The goal of integration by parts is to go from an integral $\int u(x)v'(x)\, dx$ that we don't see how to evaluate to an integral $\int v(x)u'(x)\, dx$ that we can evaluate. Generally, we choose $v'(x)$ first to be as much of the integrand as we can readily integrate; then we let $u(x)$ be the leftover part. When finding $v(x)$ from $v'(x)$, any antiderivative will work, and we usually pick the simplest one. In particular, no arbitrary constant of integration is needed in $v(x)$ because it would simply cancel out of the right-hand side of Formula (2).

Example 2: Find $\int \ln x\, dx$

Solution:
Let $u = \ln x$ and $dv = 1dx$. Then $du = (1/x)\, dx$ and $v = x$.

$$\int \ln x\, dx = \int u dv = uv - \int v du = x\ln x - \int x\left(\frac{1}{x}\right) dx = x\ln x - x + C.$$

Example 3: Evaluate (1) $\int \tan^{-1} x dx$ (2) $\int_{2/\sqrt{3}}^{2} x\sec^{-1} x dx$

Solution:

(1) Let $u = \tan^{-1}x$ and $dv = dx$. Then $du = \dfrac{dx}{1+x^2}$ and $v = x$.

$$\int \tan^{-1} x dx = \int u dv = uv - \int v du = x\tan^{-1} x - \int \frac{x}{1+x^2} dx = x\tan^{-1} x - \frac{1}{2}\int \frac{dw}{w} \quad (\text{Let } w = 1 + x^2)$$

$$= x\tan^{-1} x - \frac{1}{2}\ln|w| + C \;= x\tan^{-1} x - \frac{1}{2}\ln(1+x^2) + C\,.$$

(2) Let $u = \sec^{-1}x$ and $dv = xdx$. Then $du = \dfrac{dx}{x\sqrt{x^2-1}}$ and $v = x^2/2$.

$$\int x\sec^{-1} x dx = \int u dv = uv - \int v du = \frac{x^2}{2}\sec^{-1} x\Big]_{2/\sqrt{3}}^{2} - \int_{2/\sqrt{3}}^{2} \frac{x^2}{2x\sqrt{x^2-1}} dx \quad (\text{Let } w = x^2 - 1.)$$

$$= \frac{5\pi}{9} - \frac{\sqrt{w}}{2}\Big]_{1/3}^{3} = \frac{5\pi}{9} - \frac{1}{2}\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right) = -\frac{5\pi - 3\sqrt{3}}{9}.$$

Sometimes we have to use integration by parts more than once, as in the next example.

Example 4 Evaluate

$$\int x^2 e^x \, dx.$$

Solution: We use the integration by parts formula Equation (1) with

$$u(x) = x^2 \quad \text{and} \quad v'(x) = e^x.$$

We differentiate $u(x)$ and find an antiderivative of $v'(x)$,

$$u'(x) = 2x \quad \text{and} \quad v(x) = e^x.$$

We summarize this choice by setting $du = u'(x)\,dx$ and $dv = v'(x)\,dx$, so

$$du = 2x\,dx \quad \text{and} \quad dv = e^x\,dx.$$

We then have

$$\int \underbrace{x^2}_{u}\underbrace{e^x\,dx}_{dv} = \underbrace{x^2}_{u}\underbrace{e^x}_{v} - \int \underbrace{e^x}_{v}\underbrace{2x\,dx}_{du}.$$

Integration by parts formula

The new integral is less complicated than the original because the exponent on x is reduced by one. To evaluate the integral on the right, we integrate by parts again with $u = x$, $dv = e^x\,dx$. Then $du = dx$, $v = e^x$, and

$$\int \underbrace{x}_{u}\underbrace{e^x\,dx}_{dv} = \underbrace{x}_{u}\underbrace{e^x}_{v} - \int \underbrace{e^x}_{v}\underbrace{dx}_{du} = xe^x - e^x + C.$$

Integration by parts Equation (2)
$u = x, dv = e^x\,dx$
$v = e^x, \quad du = dx$

Using this last evaluation, we then obtain

$$\begin{aligned}\int x^2 e^x\,dx &= x^2 e^x - 2\int xe^x\,dx \\ &= x^2 e^x - 2xe^x + 2e^x + C,\end{aligned}$$

where the constant of integration is renamed after substituting for the integral on the right.

Example 5

Evaluate

$$\int e^x \cos x \, dx.$$

Solution Let $u = e^x$ and $dv = \cos x \, dx$. Then $du = e^x \, dx$, $v = \sin x$, and

$$\int e^x \cos x \, dx = e^x \sin x - \int e^x \sin x \, dx.$$

vdu

The second integral is like the first except that it has $\sin x$ in place of $\cos x$. To evaluate it, we use integration by parts with

$$u = e^x, \qquad dv = \sin x \, dx, \qquad v = -\cos x, \qquad du = e^x \, dx.$$

Then

By part again

$$\int e^x \cos x \, dx = e^x \sin x - \left(-e^x \cos x - \int (-\cos x)(e^x \, dx)\right)$$

$$= e^x \sin x + e^x \cos x - \int e^x \cos x \, dx.$$

The unknown integral now appears on both sides of the equation. Adding the integral to both sides and adding the constant of integration gives

$$2\int e^x \cos x \, dx = e^x \sin x + e^x \cos x + C_1.$$

Dividing by 2 and renaming the constant of integration gives

$$\int e^x \cos x \, dx = \frac{e^x \sin x + e^x \cos x}{2} + C. \quad \blacksquare$$

$C_1/2$

Example 6:

Evaluate $\int \tanh^{-1} 2x dx$ (2) $\int x \, (\ln x)^2 dx$.

Example 7 A Reduction Formula

Obtain a "reduction" formula that expresses the integral

$$\int \cos^n x\,dx = \int \underbrace{\cos^{n-1} x}_{u}\,\underbrace{\cos x\,dx}_{dv}$$

in terms of an integral of a lower power of $\cos x$.

Solution We may think of $\cos^n x$ as $\cos^{n-1} x \cdot \cos x$. Then we let

$$u = \cos^{n-1} x \quad \text{and} \quad dv = \cos x\,dx,$$

so that

$$du = (n-1)\cos^{n-2} x\,(-\sin x\,dx) \quad \text{and} \quad v = \sin x.$$

Hence $= \int u\,dv = uv - \int v\,du$

$$\int \cos^n x\,dx = \cos^{n-1} x \sin x + (n-1)\int \sin^2 x \cos^{n-2} x\,dx$$

change to cosine

$$= \cos^{n-1} x \sin x + (n-1)\int (1 - \cos^2 x)\cos^{n-2} x\,dx,$$

$$= \cos^{n-1} x \sin x + (n-1)\int \cos^{n-2} x\,dx - (n-1)\int \cos^n x\,dx.$$

shift to the left

If we add

$$(n-1)\int \cos^n x\,dx$$

to both sides of this equation, we obtain

$$n\int \cos^n x\,dx = \cos^{n-1} x \sin x + (n-1)\int \cos^{n-2} x\,dx.$$

We then divide through by n, and the final result is

Reduction formula

$$\int \cos^n x\,dx = \frac{\cos^{n-1} x \sin x}{n} + \frac{n-1}{n}\int \cos^{n-2} x\,dx.$$

This allows us to reduce the exponent on $\cos x$ by 2 and is a very useful formula. When n is a positive integer, we may apply the formula repeatedly until the remaining integral is either

$$\int \cos x\,dx = \sin x + C \quad \text{or} \quad \int \cos^0 x\,dx = \int dx = x + C. \qquad \blacksquare$$

Example 8:

$$\int \cos^3 x\,dx = \frac{\cos^2 x \sin x}{3} + \frac{2}{3}\int \cos x\,dx = \frac{1}{3}\cos^2 x \sin x + \frac{2}{3}\sin x + C.$$

Evaluating Definite Integrals by Parts

The integration by parts formula in (1) can be combined with Part 2 of the Fundamental Theorem in order to evaluate definite integrals by parts. Assuming that both u′ and ʋ′ are continuous over the interval [a, b], Part 2 of the Fundamental Theorem gives

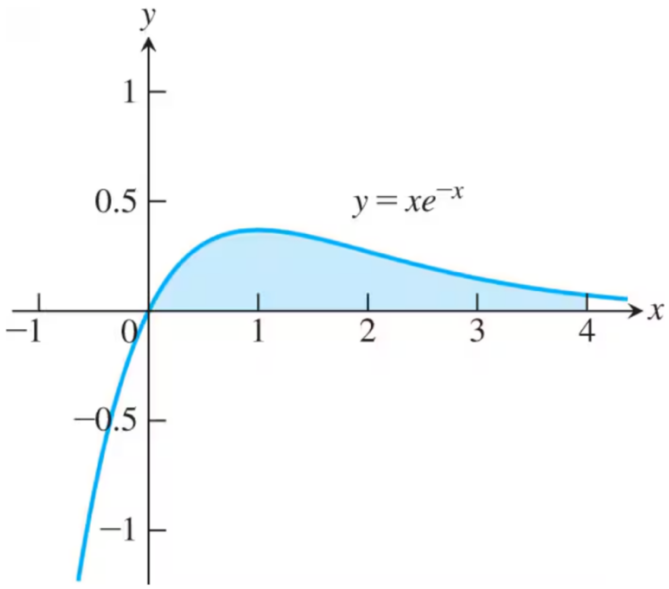
Integration by Parts Formula for Definite Integrals

$$\int_a^b u(x)\,v'(x)\,dx = u(x)\,v(x)\Big]_a^b - \int_a^b v(x)\,u'(x)\,dx \qquad (3)$$

Example 9:
Find the area of the region bounded by the curve $y = xe^{-x}$ and the x-axis from $x = 0$ to $x = 4$.

Solution
The region is shaded in Figure 8.1. Its area is $\int_0^4 xe^{-x}\,dx$.

Figure 8.1



Let $u = x$, $dv = e^{-x}dx$, $v = -e^{-x}$, and $du = dx$. Then

$$\begin{aligned}
\int_0^4 xe^{-x}\,dx &= -xe^{-x}\Big]_0^4 - \int_0^4 \left(-e^{-x}\right)\,dx \qquad \text{Integration by parts Formula (3)}\\
&= \left[-4e^{-4} - \left(-0e^{-0}\right)\right] + \int_0^4 e^{-x}\,dx\\
&= -4e^{-4} - e^{-x}\Big]_0^4\\
&= -4e^{-4} - \left(e^{-4} - e^{-0}\right) = 1 - 5e^{-4} \approx 0.91.
\end{aligned}$$

Tabular Integration

Example 10: Evaluate $\int x^4 e^{-2x} dx$.

Solution: Let $f(x) = x^4$ and $g(x) = e^{-2x}$.

$f(x)$ and its derivatives	$g(x)$ and its integrals
x^4	e^{-2x}
$4x^3$	$-\frac{1}{2}e^{-2x}$
$12x^2$	$\frac{1}{4}e^{-2x}$
$24x$	$-\frac{1}{8}e^{-2x}$
24	$\frac{1}{16}e^{-2x}$
0	$-\frac{1}{32}e^{-2x}$

$\therefore \quad \int x^4 e^{-2x} dx = -\frac{1}{4e^{2x}}\left(2x^4 + 4x^3 + 6x^2 + 6x + 3\right) + C$.

Example 11: (Substitution + By parts)
Evaluate $\int e^{\sqrt{2x-1}}\, dx.$

Solution:

Let $u = \sqrt{2x-1}.$ (Use Basic Substitution)

Then $du = \frac{2}{2\sqrt{2x-1}} dx \Rightarrow dx = u\,du$.

$\int e^{\sqrt{2x-1}} dx = \int u e^u du$ (Use by parts integration with $u = w$ and $dv = e^u du$)

$= \int w\,dv$

$= wv - \int v\,dw$

$= ue^u - \int e^u du$

$= e^{\sqrt{2x-1}}\left(\sqrt{2x-1} - 1\right) + C.$